

Complex Transseries Reversion

Geometry, Exact Convergence, Quartic Criticality,
and Sharp Borel Bounds

Research report for the ProveIt project

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Abstract

Complex transseries inversion involves several distinct questions: the existence of a formal inverse, the finiteness of its action coefficients, an ordering compatible with asymptotic magnitude, and the analytic continuation of a chosen realization. We prove a geometric and quantitative extension of the inversion framework in the ProveIt repository. For finitely many complex actions, properness of the action map with multiplicities is equivalent to a common strict decay direction. This condition gives a complete action algebra whose topology is independent of direction inside a compact decay subsector. Nevertheless, two noncollinear actions force dense changes in the ordering of the full inverse support. We classify exactly when a fixed angular ordering exists and replace it with finite angular atlases at prescribed exponential accuracy. For the actions 1 and i , the degree- N atlas has exactly $2 \sum_{n \leq N} \varphi(n) - 1$ walls and maximum angular mesh $\arctan(1/N)$.

For the inverse of $z + ae^{-z} + be^{-iz}$, we determine the exact absolute-convergence domain, including its boundary. Unequal effective amplitudes give the ordinary degree law $N^{-3/2}$; equal amplitudes give a quartic saddle and the law $N^{-5/4}$. We derive the resulting $4/3$ -power logarithmic boundary, an angular saddle bifurcation, and a uniform quartic crossover with its first correction and an $O(N^{-1})$ remainder after normalization.

A weighted local inversion theorem supplies explicit degree and action tails and preserves analytic cluster cancellation. In the Borel plane we sharpen an existing repository estimate by identifying its modified-Bessel majorant, proving the optimal universal type increase $2\sqrt{a}$, and obtaining a matching Catalan inverse-domain certificate. Applied to $\zeta^2/(\sin \pi \zeta \sin \pi \alpha \zeta)$, it gives inverse summability bounds uniform in the period parameter away from the real summation directions. A separate application of the Kamimoto–Sauzin implicit-function theorem establishes filtered continuation with an explicit path-cost set, including the delayed return of zero. Classical closure theorems are credited, the precise portions of repository questions addressed are identified, and the remaining problems at oscillatory boundaries, on later Borel sheets, and at higher transseries levels are stated explicitly.

Keywords. Transseries; series reversion; complex actions; Lagrange–Bürmann inversion; resurgence; discrete filtered sets; Stokes transport.

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1 Research scope and the results obtained

The central conclusion is that complex reversion needs a stable *filtration*, but generally cannot have one stable *ordering of every term* throughout an open sector. These requirements are often treated together. They separate even in the elementary entire function

$$F(z) = z + ae^{-z} + be^{-iz}, \quad ab \neq 0. \quad (1.1)$$

Its inverse has a convergent expansion sufficiently far into the fourth quadrant. Every generated exponential sector has a nonzero coefficient. Yet the directions where two of its exponential magnitudes agree are dense in that quadrant. Analytic inversion remains possible, and every requested finite accuracy uses only finitely many angular cells. This report proves all three statements together.

The second conclusion concerns the exact convergence boundary of (1.1). The degree sums admit a one-variable entropy maximization formula. At balanced effective amplitudes, its quadratic curvature vanishes and a quartic maximum controls the asymptotics. This gives a summable boundary law $N^{-5/4}$, a $4/3$ -power singularity in logarithmic amplitude coordinates, and a uniform critical transition as both the angle and the amplitude ratio vary. These statements are proved in [section 6](#); an explicit choice of phases also produces a cubic critical point reached by the selected inverse branch.

The third conclusion is quantitative in the Borel plane. If a forward Borel kernel has the bound $Me^{\tau|\zeta|}$, reversion of $z + \varepsilon h(z)$ admits the universal Borel bound

$$ae^{\tau r} \frac{I_1(2\sqrt{a}r)}{\sqrt{a}r}, \quad a = |\varepsilon|M, \quad r = |\zeta|. \quad (1.2)$$

The series underlying this bound, and its type estimate, already occur in the repository's negative-ray summation report [5]. We identify the Bessel function, prove sharpness, and derive matching source and target inverse half-planes. The resulting increase $2\sqrt{a}$ in the exponential-type bound is optimal over this class. A quadratic inverse attains it. This estimate gives explicit uniform control for a family whose individual real-axis residues can have arbitrarily bad small denominators.

1.1 Relationship to the repository

The repository was inspected at commit

8e9cd6f00e0ac79c4425ff6abdfa64b497cf3f41.

The relevant current directory is `Analysis/Transseries/docs/series-and-transseries/`. The canonical volume and the subsequent reports are substantial; they already contain the basic support-controlled Lagrange formulas, exact-core inversion, finite complex mixed-action coefficients, and weighted countable analytic perturbations. Those formulas are used and reproved where needed below, with no claim that they originate here. The source-level audit and its limits are recorded in [appendix A](#).

Two specific questions in *Resonance-Block Summation of Transseries* [4] motivate the extension. The question headed “Complex action geometry” in its research programme asks for cone conditions controlling complex cluster sums, products, and inverse support. The preceding inverse-resurgence question asks for continuation, local singularities, and period-uniform estimates when primitive poles become arbitrarily close. We answer the strict-cone case of the first question under stated analytic norm hypotheses. For the second, we give qualitative filtered continuation and explicit estimates away from the real summation directions. Local singularity germs on all sheets and uniform boundary estimates are left open.

1.2 What is proved, imported, or left open

Result	Status and precise boundary
Cone and coefficient criteria	Complete elementary proofs. Finite alphabets: properness with multiplicities is equivalent to a strict common half-plane; finite fibers alone require only absence of a nonnegative integer zero relation.
Direction-independent completion	Complete proof for finitely generated linear actions. Directional decay filtrations induce one completed differential algebra on compact subsectors.
Dense walls and finite atlases	Complete proofs, including an inverse realizing the obstruction, an $O(N^s)$ upper bound for s generators, and an exact quadratic wall count for actions 1, i.
Lagrange and finite Stokes transport	Established mechanisms, also present in the repository. Included with proofs and explicit weighted truncation bounds.
Exact two-action convergence	Complete proof of the full absolute-convergence criterion, including summability on its boundary and the exceptional $N^{-5/4}$ balanced degree law.
Quartic transition	Complete proofs of the $4/3$ logarithmic boundary, angle-dependent saddle bifurcation, uniform crossover, and first correction. The asymptotic statement has a uniform normalized $O(N^{-1})$ remainder.
Sharp Borel bound	Refinement of an existing repository estimate. Complete proof of sharpness and optimal universal inverse-domain constants. Applies to a kernel with a stated exponential bound.
Sine-kernel inverse continuation	Explicit application of the published Kamimoto–Sauzin theorem. The stated filtered set bounds possible singularities; it does not assert that every allowed point is singular.
General unrestricted complex transseries	No universal closure or realization theorem is claimed. Higher levels, general strong derivations, oscillatory boundaries, and critical inverse charts require additional hypotheses.

The phrase “proved here” describes the arguments in this report. It is not a claim of historical priority. The literature search supports a useful extension of the inspected repository treatment, but does not justify announcing a previously unknown general closure theorem or a field-wide breakthrough.

1.3 Established foundations

For real transseries, van der Hoeven proves compositional inverse closure and states Écalle’s Translagrange theorem [9, Theorems 5.12–5.13]. His complex-transseries treatment already makes composition sensitive to compatible directions [10, Section 2.7]. Bagayoko gives broader ordered formal inverse-closure criteria, including hyperserial settings [15, Theorem 3.1]. These are different from the sectorwise analytic problem treated here.

For resurgence, Sauzin proves nonlinear closure for a fixed closed discrete additive singular set [12]; his summability notes also establish compatibility with inversion and the alien inverse identity [11, Theorems 17.3 and 32.2]. Kamimoto and Sauzin replace a fixed discrete set by a

set filtered by continuation cost [13]. Cone completions and finite-height constructions also occur in Bridgeland’s Riemann–Hilbert framework [14, Appendix A]. The use of a filtration instead of a globally fixed order therefore has established precedents. Our specific statements concern compositional reversion, its coefficient fibers, its angular sorting obstruction, and quantitative inverse bounds.

2 Objects, conventions, and three kinds of support

Throughout, $\mathbb{N}$ contains zero. A multiindex $\nu = (\nu_1, \dots, \nu_s)$ has degree $|\nu| = \sum_j \nu_j$, factorial $\nu! = \prod_j \nu_j!$, and monomial $a^\nu = \prod_j a_j^{\nu_j}$. The symbol $F^{\circ-1}$ denotes a compositional inverse, not $1/F$. The differentiation operator D always acts on the displayed independent variable.

All logarithms and nonintegral powers are taken on an explicitly chosen branch. Equivalently, their domains can be regarded as subsets of the logarithmic cover of $\mathbb{C}^\times$. A sector in this report means a sector at infinity unless a finite point is specified. Uniform asymptotics on an angular interval are understood on a compact interval strictly inside the indicated open sector.

2.1 Formal labels, action values, and actual functions

There are three successive constructions.

1. A formal series in independent labels $X_1, \dots, X_s$ is an element of $\mathbb{C}[[X_1, \dots, X_s]]$. Its total-degree filtration makes multiplication and formal implicit inversion meaningful.
2. A choice of complex actions λ_j identifies X^ν with the action value $\Lambda_\nu = \sum_j \nu_j \lambda_j$. Coefficients with the same action must be combined. This requires finite fibers or a separate convergence argument.
3. Evaluation $X_j = e^{-\lambda_j z}$ produces an analytic function only where the corresponding sum is justified. Uniform asymptotic ordering imposes additional directional restrictions.

These steps are not interchangeable. An infinite formal coefficient sum cannot be justified by drawing a cone. A formal degree expansion cannot be declared an asymptotic expansion merely because each label has been named an exponential. Conversely, an absolutely convergent analytic sum may remain meaningful even when its support is not a Hahn support in a proposed order.

2.2 Exponential actions and Borel singularities

In an analytic exponential expansion, a linear action λ appears as $e^{-\lambda z}$, and decay on a ray $z = Re^{i\theta}$ is governed by $\operatorname{Re}(e^{i\theta}\lambda)$. In resurgence, a Borel singularity at ω can generate an exponential jump $e^{-\omega z}$ after an appropriate Laplace contour is crossed. The sets are related, but they serve different purposes. The complete Borel singularity structure may include opposite actions, even though no ray makes both corresponding exponentials decay. [Section 8.3](#) therefore uses continuation cost rather than the strict decay cone of [section 3](#).

2.3 The Borel convention

For a formal inverse-power series with no constant term, set

$$h(z) = \sum_{k \geq 0} h_k z^{-k-1}, \quad \mathcal{B}h(\zeta) = \sum_{k \geq 0} \frac{h_k}{k!} \zeta^k. \quad (2.1)$$

Thus $\mathcal{B}(h_1 h_2) = (\mathcal{B}h_1) * (\mathcal{B}h_2)$ and $\mathcal{B}(Dh) = -\zeta \mathcal{B}h$, where

$$(K_1 * K_2)(\zeta) = \int_0^\zeta K_1(t) K_2(\zeta - t) dt$$

initially uses the straight segment. Directional Laplace summation is

$$\mathcal{L}_\theta K(z) = \int_0^{e^{i\theta}\infty} e^{-z\zeta} K(\zeta) d\zeta. \quad (2.2)$$

The Borel ray angle is θ ; its associated large- z direction is close to $-\theta$. This convention prevents a sign ambiguity when discussing decay cones and summation directions in the same article.

3 The sharp cone criterion and the completed action algebra

Fix nonzero complex numbers $\lambda_1, \dots, \lambda_s$, and write

$$A : \mathbb{N}^s \longrightarrow \mathbb{C}, \quad A(\nu) = \Lambda_\nu := \sum_{j=1}^s \nu_j \lambda_j, \quad P = A(\mathbb{N}^s). \quad (3.1)$$

The word “proper” below includes multiplicities: the preimage of every bounded subset of $\mathbb{C}$ must be finite. This is stronger than local finiteness of the distinct-value set P .

Theorem 3.1 (Finite-action cone criterion). *The following conditions are equivalent:*

- (i) *there is a unit complex number u with $\operatorname{Re}(u\lambda_j) > 0$ for every j ;*
- (ii) $0 \notin \operatorname{conv}\{\lambda_1, \dots, \lambda_s\}$;
- (iii) *the action map A is proper;*
- (iv) *there is $c > 0$ such that $|\Lambda_\nu| \geq c|\nu|$ for every ν .*

Under these conditions the optimal directional margin is

$$\max_{|u|=1} \min_j \operatorname{Re}(u\lambda_j) = \operatorname{dist}(0, \operatorname{conv}\{\lambda_1, \dots, \lambda_s\}) > 0. \quad (3.2)$$

Proof. A strictly positive linear functional separates the convex hull from zero, so (i) implies (ii). Conversely, let p be the point of that compact convex hull closest to zero. If (ii) holds, $p \neq 0$. Differentiating squared distance along the segment from p to λ_j gives

$$\operatorname{Re}(\bar{p}(\lambda_j - p)) \geq 0.$$

For $u = \bar{p}/|p|$ this implies $\operatorname{Re}(u\lambda_j) \geq |p|$. For any other unit u , since p is a convex combination of the generators,

$$\min_j \operatorname{Re}(u\lambda_j) \leq \operatorname{Re}(up) \leq |p|.$$

This proves both (i) and (3.2).

A positive margin gives $|\Lambda_\nu| \geq \operatorname{Re}(u\Lambda_\nu) \geq c|\nu|$, so (i) implies (iv). Condition (iv) plainly implies (iii). To prove the remaining implication, suppose 0 lies in the convex hull. Choose $t_j \geq 0$ with $\sum_j t_j = 1$ and $\sum_j t_j \lambda_j = 0$, and set $\nu_j(k) = \lfloor kt_j \rfloor$. Then

$$|\nu(k)| \geq k - s, \quad |\Lambda_{\nu(k)}| \leq \sum_j |\lambda_j|.$$

Infinitely many different multiindices therefore map to a bounded set. This contradicts (iii). $\square$

Proposition 3.2 (Finite fibers are a weaker condition). *Every fiber of A is finite if and only if there is no nonzero $\mathbf{k} \in \mathbb{N}^s$ with $\Lambda_{\mathbf{k}} = 0$.*

Proof. A nonzero such $\mathbf{k}$ gives the infinite fiber $A^{-1}(0) \supset \{n\mathbf{k} : n \geq 0\}$. Conversely, Dickson's lemma says that an infinite subset of $\mathbb{N}^s$ contains two distinct componentwise comparable elements. Apply it to an infinite fiber: if $\boldsymbol{\nu} \geq \boldsymbol{\mu}$ and $A(\boldsymbol{\nu}) = A(\boldsymbol{\mu})$, then $\mathbf{k} = \boldsymbol{\nu} - \boldsymbol{\mu} \neq 0$ is a nonnegative integer zero relation. For completeness, the needed form of Dickson's lemma follows by induction on s : an infinite sequence of first coordinates has an infinite nondecreasing subsequence, and repeating this for the remaining coordinates leaves two comparable terms. $\square$

Example 3.3 (Two distinct failures). For $\lambda_1 = 1$, $\lambda_2 = -1$, the set $P = \mathbb{Z}$ is discrete, but every action has infinitely many representations. For $\lambda_1 = 1$, $\lambda_2 = -\sqrt{2}$, the map A is injective, because a rational relation with $\sqrt{2}$ is impossible. Nevertheless it is not proper and there is no common decay direction. The rounding construction in the preceding proof already gives infinitely many bounded actions; rational approximation also gives nonzero actions approaching zero. Thus exclusion of exact integer resonance does not replace strict cone geometry.

3.1 Uniform directional weights

Let J be a connected open angular interval on which $d_j(\theta) = \operatorname{Re}(e^{i\theta}\lambda_j) > 0$ for all j . We call J a *decay chamber*. Its width is at most π . Fix a compact interval $I \subset J$, and put

$$c_I = \min_{\theta \in I, j} d_j(\theta) > 0, \quad L = \max_j |\lambda_j|, \quad w_I(\Lambda) = \min_{\theta \in I} \operatorname{Re}(e^{i\theta}\Lambda). \quad (3.3)$$

For any representation $\Lambda = \Lambda_{\boldsymbol{\nu}}$,

$$c_I|\boldsymbol{\nu}| \leq w_I(\Lambda) \leq \operatorname{Re}(e^{i\theta}\Lambda) \leq L|\boldsymbol{\nu}|. \quad (3.4)$$

The weight w_I is superadditive: $w_I(\Lambda + \Gamma) \geq w_I(\Lambda) + w_I(\Gamma)$.

Theorem 3.4 (A common completion across the chamber). *Under the cone condition, formal sums*

$$\mathcal{A}_P = \left\{ \sum_{\Lambda \in P} c_{\Lambda} E_{\Lambda} : c_{\Lambda} \in \mathbb{C} \right\}, \quad E_{\Lambda} E_{\Gamma} = E_{\Lambda + \Gamma},$$

form a complete filtered commutative algebra for the filtration

$$\mathcal{F}_I^T = \{f : c_{\Lambda} = 0 \text{ whenever } w_I(\Lambda) < T\}.$$

Every multiplication coefficient is a finite sum. The derivation $DE_{\Lambda} = -\Lambda E_{\Lambda}$ is continuous. For every $\theta \in I$, the directional weight $\operatorname{Re}(e^{i\theta}\Lambda)$ induces the same completion and the same topology as w_I .

Proof. Only finitely many multiindices have $c_I|\boldsymbol{\nu}| < T$, so only finitely many actions have $w_I(\Lambda) < T$. If $\Lambda + \Gamma = \Sigma$, choose representations for both summands. Their sum represents Σ and has degree at most $\operatorname{Re}(e^{i\theta}\Sigma)/c_I$. There are only finitely many such pairs. This proves that convolution of coefficients is defined. Superadditivity proves $\mathcal{F}_I^T \mathcal{F}_I^U \subseteq \mathcal{F}_I^{T+U}$.

A Cauchy net for this filtration stabilizes every finite weight truncation; the stabilized coefficients give its unique limit. Differentiation leaves the action support unchanged, so it is continuous. Finally, for $\theta, \phi \in I$ and any representation of Λ , (3.4) gives

$$\frac{c_I}{L} \operatorname{Re}(e^{i\phi}\Lambda) \leq \operatorname{Re}(e^{i\theta}\Lambda) \leq \frac{L}{c_I} \operatorname{Re}(e^{i\phi}\Lambda). \quad (3.5)$$

The same comparison applies to w_I . Cofinal truncation thresholds therefore define identical topologies. $\square$

There is also a map, continuous for the adic and weight topologies,

$$\mathbb{C}[[X_1, \dots, X_s]] \longrightarrow \mathcal{A}_P, \quad X^\nu \longmapsto E_{\Lambda_\nu}.$$

The coefficient at Λ is the finite sum over $A^{-1}(\Lambda)$. Its kernel is the closure of the ideal of binomials $X^\nu - X^\mu$ for which $\Lambda_\nu = \Lambda_\mu$: this follows by grouping each finite weight truncation into its fibers. Thus exact resonances are handled by a continuous quotient, without choosing one angular ordering.

Corollary 3.5 (Formal inverse independent of the sorting direction). *For $F(z) = z + \sum_j a_j E_{\lambda_j}$ there is a unique correction in the positive-weight ideal solving*

$$v + \sum_j a_j E_{\lambda_j} \exp(-\lambda_j v) = 0. \quad (3.6)$$

It is the same element of $\mathcal{A}_P$ for every directional filtration with angle in I .

Proof. The exponential is defined by its power series in the positive-weight element v . Starting at $v_0 = 0$, iterate the right side of $v = -\sum_j a_j E_{\lambda_j} \exp(-\lambda_j v)$. If two inputs agree below weight T , their outputs agree below weight $T + c_I$. Iteration therefore converges in the complete filtration and gives uniqueness by the same improvement. The equation and the topology are independent of the sorting direction. $\square$

This is the algebraic part of the proposed repair. The following sections supply analytic evaluation, error estimates, and the finite angular orderings needed for computation.

4 Weighted inversion in an exact core coordinate

The coefficient formula in this section is classical Lagrange–Bürmann, and its exact-core and mixed-action forms are already present in the repository [1, 2, 3]. We record a weighted version to make action cutoffs and cluster limits explicit.

Suppose F_0 has a holomorphic inverse branch G_0 on a domain U , and

$$F = F_0 + h, \quad H = h \circ G_0.$$

Writing $\xi = F_0(z)$ reduces $F(z) = w$ to

$$\xi + H(\xi) = w, \quad z = G_0(\xi). \quad (4.1)$$

No expansion of G_0 is required. In particular, logarithmic drift can be kept in the core rather than repeatedly Taylor-expanded through a large exponential.

Theorem 4.1 (Local inversion with degree and action tails). *Let G_0 and each H_j , j in a finite or countable index set, be holomorphic on a neighborhood of the closed disk $\overline{D}(w, r)$. Let $p_j > 0$, $\sigma \geq 0$, and suppose*

$$\sum_j e^{\sigma p_j} \|H_j\|_{\overline{D}(w, r)} \leq \kappa r, \quad 0 < \kappa < 1, \quad \|G'_0\|_{\overline{D}(w, r)} \leq C. \quad (4.2)$$

Put $H = \sum_j H_j$. Then (4.1) has exactly one root ξ in $D(w, r)$, counted with multiplicity, and the selected inverse is

$$G(w) = G_0(w) + \sum_{n \geq 1} \frac{(-1)^n}{n!} D^{n-1}(G'_0(w) H(w)^n). \quad (4.3)$$

The multigraded form is

$$G(w) - G_0(w) = \sum_{\substack{\boldsymbol{\nu} \in \mathbb{N}^{(J)} \\ |\boldsymbol{\nu}| \geq 1}} T_{\boldsymbol{\nu}}(w), \quad T_{\boldsymbol{\nu}} = \frac{(-1)^{|\boldsymbol{\nu}|}}{\boldsymbol{\nu}!} D^{|\boldsymbol{\nu}|-1} \left(G'_0 \prod_j H_j^{\nu_j} \right), \quad (4.4)$$

where $\mathbb{N}^{(J)}$ means finite-support multiindices. With $p(\boldsymbol{\nu}) = \sum_j \nu_j p_j$, one has

$$\sum_{|\boldsymbol{\nu}|=n} e^{\sigma p(\boldsymbol{\nu})} |T_{\boldsymbol{\nu}}(w)| \leq \frac{Cr}{n} \kappa^n, \quad (4.5)$$

$$\sum_{|\boldsymbol{\nu}| > N} |T_{\boldsymbol{\nu}}(w)| \leq \frac{Cr \kappa^{N+1}}{(N+1)(1-\kappa)}, \quad (4.6)$$

$$\sum_{p(\boldsymbol{\nu}) > B} |T_{\boldsymbol{\nu}}(w)| \leq Cr [-\log(1-\kappa)] e^{-\sigma B}. \quad (4.7)$$

For the last estimate a useful action tail requires $\sigma > 0$. All series are locally normally convergent wherever the hypotheses hold with locally uniform constants.

Proof. The series for H converges uniformly on the closed disk and is holomorphic in its interior. For $|t| < 1/\kappa$, choose $|t|\kappa r < \rho < r$. On the circle of radius ρ , $|tH(\xi)| \leq |t|\kappa r < |\xi - w|$, so Rouché's theorem gives exactly one zero of $\xi + tH(\xi) - w$, with multiplicity one. Every zero in $D(w, r)$ satisfies $|\xi - w| \leq |t|\kappa r$, so it is the unique zero in the whole disk. The selected root and its image under G_0 are holomorphic in t on a disk containing $t = 1$.

Here is a direct derivation of the coefficient identity. For any holomorphic A on the disk, the residue formula for the unique root gives

$$A(\xi(t)) = \frac{1}{2\pi i} \int_{|\zeta-w|=\rho} A(\zeta) \frac{1 + tH'(\zeta)}{\zeta - w + tH(\zeta)} d\zeta.$$

Fix such an interior circle and expand near $t = 0$, where the geometric expansion converges on it. For $n \geq 1$, its t^n coefficient is the sum of

$$\frac{(-1)^n}{n!} D^n(AH^n) \quad \text{and} \quad \frac{(-1)^{n-1}}{(n-1)!} D^{n-1}(AH'H^{n-1}).$$

The product rule combines these as $(-1)^n D^{n-1}(A'H^n)/n!$. Taking $A = G_0$ gives (4.3). For finitely many H_j , the multinomial theorem gives (4.4).

Write $h_j = \|H_j\|_{\overline{D}(w,r)}$. Apply Cauchy's derivative estimate on circles of radius $\rho < r$ and let $\rho \uparrow r$. This gives, for $n = |\boldsymbol{\nu}|$,

$$|T_{\boldsymbol{\nu}}(w)| \leq \frac{C(n-1)!}{r^{n-1} \boldsymbol{\nu}!} \prod_j h_j^{\nu_j}.$$

Multiplying by $e^{\sigma p(\boldsymbol{\nu})}$ and summing at degree n yields (4.5). Its sum over n is finite; consequently the countable formula follows by finite truncation and normal convergence. Summing κ^n/n for $n > N$ proves (4.6). For $p(\boldsymbol{\nu}) > B$, remove the factor $e^{\sigma p(\boldsymbol{\nu})} \geq e^{\sigma B}$ and sum all degrees. This proves (4.7). $\square$

4.1 General holomorphic phases

The theorem does not require linear phases. Suppose $H_j = a_j e^{-\Phi_j}$, with all functions holomorphic on the inversion disk. For $\Phi_\nu = \sum_j \nu_j \Phi_j$ the mixed term is

$$T_\nu = \frac{(-1)^{|\nu|}}{\nu!} e^{-\Phi_\nu} (D - \Phi'_\nu)^{|\nu|-1} \left(G'_0 \prod_j a_j^{\nu_j} \right). \quad (4.8)$$

Indeed $D(e^{-\Phi} A) = e^{-\Phi} (D - \Phi') A$, and induction proves the formula. The operator power is an iterated differential operator; when $\Phi'' \neq 0$, it must not be expanded as a binomial with constant coefficients.

Equation (4.8) preserves the *phase labels* under addition. If derivatives of the phases contain other exponential scales, those scales live in the coefficient algebra and may have to be promoted into a larger transseries alphabet. Thus this analytic theorem does not assert closure of every proposed minimal formal alphabet.

4.2 Cluster cancellation and perturbation stability

Each H_j can be an already grouped cluster, for example a contour-defined sum whose individual residue terms become large as poles coalesce. The hypothesis of [theorem 4.1](#) is a norm of the whole analytic block. It does not require a sum of the absolute values of its individual residues. Once these block norms are proved, no inverse coefficient introduces a denominator involving a difference of action values.

Proposition 4.2 (Stability of the selected inverse). *Suppose H, K are holomorphic on $\overline{D}(w, r)$ and both have norm at most $\kappa r < r$. Use the same G_0 , with derivative norm at most C . The selected inverses satisfy*

$$|G_H(w) - G_K(w)| \leq \frac{C \|H - K\|_{\overline{D}(w, r)}}{1 - \kappa}. \quad (4.9)$$

Consequently inversion commutes with any locally uniform cluster limit for which the disk constants remain uniform.

Proof. Subtract (4.3) for H and K . On the disk,

$$|H^n - K^n| \leq n(\kappa r)^{n-1} \|H - K\|.$$

Cauchy's estimate bounds the difference of the degree- n terms by $C\kappa^{n-1} \|H - K\|$. Sum the geometric series. Locally uniform convergence, including parameter dependence, follows from the same estimate. $\square$

This is a transport theorem for proved cluster estimates. It does not construct complex Borel clusters or supply their missing norm bounds. That distinction is material to the repository question addressed here.

4.3 When an action cutoff is actually finite

For countably many primitive blocks, assume in addition that $\{j : p_j \leq B\}$ is finite for every B . Then the weights have a positive minimum, and only finitely many multiindices have $p(\nu) \leq B$. Under this condition (4.7) is a finite computation certificate. Without it, the weighted analytic sum is still legitimate, but an action cutoff may contain infinitely many terms.

A common half-plane by itself does not cure this countable issue. The actions $1 + 1/j$, $j \geq 2$, all have a uniform positive decay margin but infinitely many lie in a bounded action interval. Taking both $1 + 1/j$ and $1 - 1/j$ gives infinitely many two-letter representations of the action 2. Summable amplitudes can still make the analytic regrouping converge. They cannot make the corresponding unrestricted formal coefficient extraction a finite sum.

5 Convergent inverse sectors and dense angular walls

5.1 An explicit inverse and its remainder

Consider the finite exponential perturbation

$$F(z) = z + \sum_{j=1}^s a_j e^{-\lambda_j z}, \quad L = \max_j |\lambda_j|, \quad S(w) = \sum_j |a_j e^{-\lambda_j w}|. \quad (5.1)$$

Theorem 5.1 (Convergent action expansion). *If $Q(w) = eLS(w) < 1$, the root of $F(z) = w$ with $|z - w| < 1/L$ is unique and simple. It is given by the absolutely, locally normally convergent sum*

$$G(w) = w - \sum_{\substack{\nu \in \mathbb{N}^s \\ |\nu| \geq 1}} \frac{\Lambda_\nu^{|\nu|-1}}{\nu!} a^\nu e^{-\Lambda_\nu w}. \quad (5.2)$$

At degree one, Λ^0 means 1, including if a zero action occurs in a more general version of the formula. Truncation to $|\nu| \leq N$ satisfies both the disk estimate (4.6) with $C = 1$, $r = 1/L$, $\kappa = Q$, and the sharper bound

$$|G(w) - G_{\leq N}(w)| \leq \frac{Q(w)^{N+1}}{L\sqrt{2\pi}(N+1)^{3/2}(1-Q(w))}. \quad (5.3)$$

The convergence statement does not require a cone. A cone makes its degree filtration uniformly asymptotic at infinity.

Proof. On $|z - w| \leq 1/L$, the perturbation is bounded by $eS(w) < 1/L$. Apply theorem 4.1 with $G_0 = \text{id}$. Differentiating the degree- n multinomial term multiplies it by $(-\Lambda_\nu)^{n-1}$, which combines with $(-1)^n$ to give the minus sign in (5.2). The degree- n absolute sum is at most

$$\frac{(Ln)^{n-1}}{n!} S(w)^n \leq \frac{Q(w)^n}{L\sqrt{2\pi} n^{3/2}}, \quad (5.4)$$

using $n! \geq \sqrt{2\pi n}(n/e)^n$. Summing the geometric majorant from $n = N + 1$ proves (5.3). $\square$

For one action the inverse is $G(w) = w + \lambda^{-1}W_0(-\lambda a e^{-\lambda w})$ near zero amplitude. The branch point of W_0 at $-1/e$ shows that the universal threshold $LS < 1/e$ cannot be improved for all finite alphabets. Particular alphabets can have substantially larger convergence domains; the next section determines one of them exactly.

5.2 The obstruction to a fixed all-orders ordering

Definition 5.2. A *dominance wall* for distinct actions Λ, Γ is a direction θ satisfying

$$\text{Re}(e^{i\theta}(\Lambda - \Gamma)) = 0. \quad (5.5)$$

It is a comparison of exponential rates. It is not, by definition, a Borel singularity or a Stokes jump.

On a ray $w = Re^{i\theta}$, the modulus ratio is $\exp[-R \operatorname{Re}(e^{i\theta}(\Lambda - \Gamma))]$. A sign change in the rate difference therefore reverses the asymptotic comparison.

Theorem 5.3 (Classification of stable angular orderings). *Suppose a finite nonzero action family admits a decay chamber. The magnitude ordering of the distinct monomials in its full additive semigroup is independent of angle on some nonempty open subinterval if and only if all generators lie on a single positive real ray.*

If two generators are not real-collinear, the comparisons of their positive integer multiples alone have dense dominance walls in every decay chamber. This obstruction occurs in the actual convergent inverse of $z + ae^{-\lambda_1 z} + be^{-\lambda_2 z}$ when $ab \neq 0$.

Proof. If $\lambda_j = t_j \lambda_0$ with $t_j > 0$, comparison reduces to comparison of the real sums $\sum_j \nu_j t_j$, multiplied by the common positive factor $\operatorname{Re}(e^{i\theta} \lambda_0)$.

Otherwise choose noncollinear λ_1, λ_2 . Put $d_j(\theta) = \operatorname{Re}(e^{i\theta} \lambda_j) > 0$. Direct differentiation gives

$$\left(\frac{d_1}{d_2}\right)' = \frac{\operatorname{Im}(\overline{\lambda_1} \lambda_2)}{d_2^2} \neq 0. \quad (5.6)$$

Every open angular interval therefore maps onto an open interval of positive ratios. The positive rationals are dense, and $d_1/d_2 = p/q$ is precisely the wall comparing $q\lambda_1$ and $p\lambda_2$. The ratio crosses that value strictly.

For two noncollinear actions the map $(n, m) \mapsto n\lambda_1 + m\lambda_2$ is injective. Its nonzero values cannot vanish. Every coefficient

$$-\frac{(n\lambda_1 + m\lambda_2)^{n+m-1} a^n b^m}{n!m!}$$

in (5.2) is nonzero. Hence the obstruction is realized by the actual inverse support, not by a formal overestimate. $\square$

The theorem does not invalidate formal complex transseries with a chosen order. It shows that such an order cannot agree with all asymptotic magnitude comparisons throughout an open interval in this example. Theorem 3.4 remains valid: the filtration topology is stable while the order inside finite truncations changes.

5.3 A finite atlas at prescribed exponential accuracy

Fix $I \subset J$ compact, set $c = c_I$, and let $A_0 = \sum_j |a_j|$. Then $S(Re^{i\theta}) \leq A_0 e^{-cR}$. Choose R_0 with $Q_0 = eLA_0 e^{-cR_0} < 1$. Formula (5.3) gives

$$|G - G_{\leq N}| \leq \frac{(eLA_0)^{N+1} e^{-(N+1)cR}}{L\sqrt{2\pi}(N+1)^{3/2}(1-Q_0)}, \quad R \geq R_0, \quad \theta \in I. \quad (5.7)$$

For requested exponential precision e^{-TR} , choose

$$N = \left\lceil \frac{T}{c} \right\rceil. \quad (5.8)$$

Since $(N+1)c > T$, the error is $o(e^{-TR})$, uniformly on I . Moreover every action of rate at most T belongs to degree at most N .

Proposition 5.4 (Finite atlas and its size). *Group exact resonances among $\{\Lambda_\nu : 1 \leq |\nu| \leq N\}$ and subdivide I at their dominance walls. The retained exponential rates have one fixed strict ordering on each open cell. There are at most*

$$\frac{(2N+1)^s - 1}{2} \quad (5.9)$$

candidate walls, and hence at most one more than this many cells. If the set of rates below T must be fixed on each cell, additionally subdivide at $\text{Re}(e^{i\theta}\Lambda) = T$; this adds at most twice the number of retained actions. For fixed actions and I , both bounds are $O(T^s)$ under (5.8).

Proof. Every action difference is represented by an integer vector in $[-N, N]^s$. Up to sign there are at most (5.9) nonzero such vectors. A nonzero complex difference has at most one wall in an interval of width less than π . Vectors mapping to zero are exact resonances and are omitted. Each threshold equation is a sinusoidal equation and has at most two solutions in the chamber. $\square$

These are existence and combinatorial bounds. Implementing them for arbitrary complex parameters requires exact equality and ordering information, or certified enclosures proving the required comparisons. No bit-complexity claim for unspecified computable real numbers follows from the absence of a separation constant in the mathematical bound.

5.4 Exact wall count for two orthogonal actions

For $\lambda_1 = 1$, $\lambda_2 = i$, the chamber is $-\pi/2 < \theta < 0$. The inverse is

$$G(w) = w - \sum_{\substack{m, n \geq 0 \\ m+n \geq 1}} \frac{(m+in)^{m+n-1} a^m b^n}{m!n!} e^{-(m+in)w}. \quad (5.10)$$

The walls of the truncation to total degree at most N are exactly

$$\Theta_N = \{-\arctan(p/q) : 1 \leq p, q \leq N, \gcd(p, q) = 1\}. \quad (5.11)$$

Indeed an interior wall requires the real and imaginary parts of an action difference to have opposite signs. Both absolute values are at most N . Conversely the retained pure actions p and iq give every ratio in (5.11).

Theorem 5.5 (Exact quadratic atlas). *For $N \geq 1$, the number of distinct interior walls is*

$$W_N = 2 \sum_{k=1}^N \varphi(k) - 1 = \frac{6}{\pi^2} N^2 + O(N \log N). \quad (5.12)$$

After the two quadrant boundaries are added, the largest angular cell has width exactly $\arctan(1/N)$.

Proof. Count coprime pairs in $[1, N]^2$ on the two sides of the diagonal; the only coprime diagonal point is $(1, 1)$. This gives the first identity. Equivalently the count is $\sum_{d \leq N} \mu(d) \lfloor N/d \rfloor^2$. Replacing the square by $(N/d)^2 + O(N/d)$ and using $\sum_{d \geq 1} \mu(d)/d^2 = 1/\zeta(2) = 6/\pi^2$ gives the asymptotic.

For the mesh, on the half-quadrant with slope between zero and one the wall set includes all slopes p/N , reduced if necessary. Concavity of $\arctan$ bounds every associated angular gap by $\arctan(1/N)$. Reciprocal symmetry handles the other half. Equality holds at the boundary, because no positive fraction with numerator and denominator at most N is smaller than $1/N$. $\square$

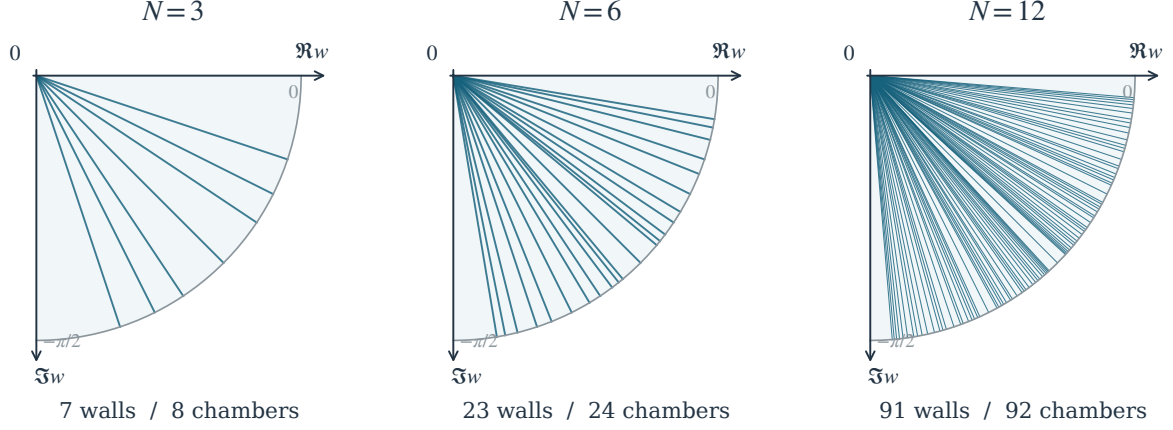


Figure 1: Finite rate-ordering atlases for the two actions 1 and i. The degree cutoffs 3, 6, 12 have respectively 7, 23, 91 interior walls. The all-orders wall set is dense, while each finite cutoff is explicit. These lines mark changes of exponential-rate ordering, not Stokes jumps.

On $I_\eta = [-\pi/2 + \eta, -\eta]$, $0 < \eta < \pi/4$, one may take $c = \sin \eta$. Precision e^{-TR} therefore needs at most $O(T^2)$ cells. The full-quadrant count is exactly quadratic; no lower bound for every possible restricted interval is needed here.

Polynomial or logarithmic amplitudes change the comparison of full term magnitudes by lower-order contributions. For example a power factor can shift a simple wall by $O(\log R/R)$. The theorem above orders *exponential rates*. Uniform control of such moving, amplitude-sensitive walls is a separate extension discussed below.

6 The exact two-action convergence domain and its quartic boundary

The preceding certificate is uniform over all finite alphabets. It loses information when complex action vectors partially cancel inside $\Lambda_\nu^{|\nu|-1}$. For the orthogonal pair 1, i we can retain that information and determine the full domain of absolute convergence. The coefficient formula and the saddle method are classical; the exact calculation below is the model-specific result.

Set $A = |ae^{-iw}|$, $B = |be^{-iw}|$, and define the absolute contribution of total degree N by

$$D_N(A, B) = \sum_{m+n=N} \frac{|m + in|^{N-1}}{m!n!} A^m B^n. \quad (6.1)$$

For $0 < p < 1$, put

$$\begin{aligned} r(p) &= \sqrt{p^2 + (1-p)^2}, \\ H(p) &= -p \log p - (1-p) \log(1-p), \\ \Psi_{A,B}(p) &= 1 + \log r(p) + H(p) + p \log A + (1-p) \log B, \end{aligned} \quad (6.2)$$

$$\mathcal{R}(A, B) = \exp \left(\max_{0 \leq p \leq 1} \Psi_{A,B}(p) \right). \quad (6.3)$$

Endpoint values are defined by continuity. The notation $\mathcal{R}$ denotes a coefficient growth factor, not a radius in the w -plane.

Theorem 6.1 (Sharp convergence and boundary exponents). *For $A, B > 0$, the inverse series (5.10) converges absolutely if and only if $\mathcal{R}(A, B) \leq 1$. Its maximizing proportion p_* is unique, lies in $(0, 1)$, and satisfies*

$$\log \frac{A}{B} + \log \frac{1-p_*}{p_*} + \frac{2p_*-1}{p_*^2 + (1-p_*)^2} = 0. \quad (6.4)$$

For $A \neq B$,

$$D_N(A, B) \sim \frac{r(p_*)}{\sqrt{2\pi}|2p_*-1|} \mathcal{R}(A, B)^N N^{-3/2}. \quad (6.5)$$

For $A = B = q > 0$,

$$\mathcal{R}(q, q) = e\sqrt{2}q, \quad D_N(q, q) \sim C_4(e\sqrt{2}q)^N N^{-5/4}, \quad C_4 = \frac{3^{1/4}\Gamma(1/4)}{2\sqrt{2}\pi}. \quad (6.6)$$

Thus the symmetric boundary is exactly $q = 1/(e\sqrt{2})$, and absolute convergence holds on that boundary.

Proof. Differentiation gives

$$\Psi''_{A,B}(p) = -\frac{(2p-1)^2}{p(1-p)(p^2 + (1-p)^2)^2}. \quad (6.7)$$

It is negative except at $p = 1/2$. Hence Ψ' is strictly decreasing; its endpoint limits are $+\infty$ and $-\infty$. This proves the existence and uniqueness of the maximizer and (6.4). The maximizer is $1/2$ exactly when $A = B$.

For $p = m/N$ in a compact subinterval of $(0, 1)$, uniform Stirling asymptotics express the corresponding summand of (6.1) as

$$\frac{\exp(N\Psi_{A,B}(p))}{2\pi N^2 r(p) \sqrt{p(1-p)}} (1 + O(N^{-1})). \quad (6.8)$$

The endpoints are harmless: $\binom{N}{m} \leq \exp(NH(p))$ and the lower Stirling bound for $N!$ bound every summand by $CN^{-3/2} \exp(N\Psi_{A,B}(p))$, uniformly even at $p = 0, 1$. Any closed set separated from p_* has an exponent smaller by a positive constant and contributes exponentially little.

If $p_* \neq 1/2$, put $p = p_* + uN^{-1/2}$. The exponent difference tends to $\Psi''(p_*)u^2/2$, and the rescaled lattice spacing is $N^{-1/2}$. A local quadratic upper bound supplies Gaussian domination. The resulting Riemann sum yields

$$D_N \sim \frac{\mathcal{R}^N N^{-3/2}}{\sqrt{2\pi} r(p_*) \sqrt{p_*(1-p_*)} \sqrt{-\Psi''(p_*)}}.$$

Substitution of (6.7) gives (6.5).

If $A = B = q$, write $p = 1/2 + x$. Direct expansion gives

$$\Psi_{q,q}(1/2 + x) = 1 + \log(\sqrt{2}q) - \frac{16}{3}x^4 + \frac{128}{15}x^6 + O(x^8). \quad (6.9)$$

The quadratic term cancels. Set $x = uN^{-1/4}$; the lattice spacing in u is $N^{-3/4}$ and the prefactor in (6.8), without N^{-2} , tends to $\sqrt{2}/\pi$. A quartic upper bound near zero and the previous exponential estimate away from zero justify dominated Riemann-sum convergence. Therefore

$$D_N \sim \mathcal{R}^N N^{-5/4} \frac{\sqrt{2}}{\pi} \int_{\mathbb{R}} \exp(-16u^4/3) du.$$

Substituting $v = (16/3)u^4$ evaluates this integral and gives C_4 . For $\mathcal{R} > 1$ the degree sum fails to tend to zero; for $\mathcal{R} < 1$ it decays geometrically. At $\mathcal{R} = 1$, the exponents $3/2$ and $5/4$ are both summable. This proves the criterion. $\square$

If one of A, B is zero, the one-action criterion is respectively $eB \leq 1$ or $eA \leq 1$. At equal nonzero moduli the exact criterion improves the general sufficient condition $e(A + B) < 1$ by a factor $\sqrt{2}$ in the permitted common modulus.

On $\mathcal{R} < 1$, the series is holomorphic in its two effective amplitude variables and continues the inverse germ at zero amplitude. The domain is connected, since scaling both moduli by $t > 0$ scales $\mathcal{R}$ by t . Its inverse equation holds first near zero and then throughout the domain by analyticity. At $\mathcal{R} = 1$, absolute convergence gives a continuous radial boundary value still satisfying that equation. It need not give a holomorphic inverse there.

6.1 A 4/3-power law for the convergence boundary

Write $g = (\log A + \log B)/2$ and $\delta = \log(A/B)$. The optimized function $\log \mathcal{R}$ is convex in logarithmic moduli, because it is a maximum of affine functions of those moduli.

Corollary 6.2 (Boundary regularity). *As $\delta \rightarrow 0$,*

$$\log \mathcal{R} = 1 + g + \frac{1}{2} \log 2 + \frac{3^{4/3}}{16} |\delta|^{4/3} + \frac{3}{160} \delta^2 + O(|\delta|^{8/3}). \quad (6.10)$$

Consequently the boundary $\mathcal{R} = 1$, expressed as $g(\delta)$, is C^1 but not C^2 at zero.

Proof. The local optimization is $\max_x (\delta x - 16x^4/3 + 128x^6/15 + O(x^8))$. Its maximizer satisfies $x = (3\delta/64)^{1/3} + O(\delta)$, with the real cube root. More precisely, put $t = \text{sgn}(\delta)|\delta|^{1/3}$ and $x = ty$. The critical equation divided by t^3 has a nonzero derivative in y at $t = 0$, $y = 3^{1/3}/4$. The analytic implicit-function theorem gives an expansion of y in t^2 . The leading quartic optimum is $3^{4/3}|\delta|^{4/3}/16$. The x^6 term evaluated at its leading maximizer contributes $3\delta^2/160$. Displacement of that maximizer changes the value only at order $|\delta|^{8/3}$, since the leading part is stationary. This proves (6.10). The derivative of the optimized part in δ equals the unique maximizing x , which behaves as a nonzero multiple of $\text{sgn}(\delta)|\delta|^{1/3}$. The regularity statement follows. $\square$

This 4/3 exponent describes the graph of a convergence boundary. It does not identify that graph with the discriminant of a general cusp catastrophe.

6.2 The transition as the action angle varies

Rotate two unit actions to $\lambda_+ = e^{i\beta/2}$, $\lambda_- = e^{-i\beta/2}$, where $0 < \beta < \pi$. Both remain in a strict cone. For equal effective moduli q , replace $r(p)$ by

$$\begin{aligned} r_\beta(p) &= |p\lambda_+ + (1-p)\lambda_-|, \\ r_\beta(p)^2 &= c^2(1 + t^2(2p-1)^2), \quad c = \cos(\beta/2), \quad t = \tan(\beta/2). \end{aligned} \quad (6.11)$$

The degree growth is governed by $\log r_\beta + H$.

Proposition 6.3 (Angular saddle transition). *For $0 < \beta < \pi/2$, the unique maximizing proportion is $1/2$ and*

$$D_N(q, q; \beta) \sim \frac{(2eq \cos(\beta/2))^N}{\sqrt{2\pi \cos \beta} N^{3/2}}. \quad (6.12)$$

At $\beta = \pi/2$ the maximum is quartic and (6.6) applies. For $\pi/2 < \beta < \pi$, exactly two nondegenerate proportions maximize the exponent. Their distances from $1/2$ are $y_*/2$, where $y_* \in (0, 1)$ uniquely solves

$$\frac{\operatorname{artanh} y_*}{y_*} = \frac{t^2}{1 + t^2 y_*^2}. \quad (6.13)$$

Both contribute ordinary $N^{-3/2}$ terms.

Proof. In the coordinate $y = 2p - 1$, the derivative of $\log r_\beta + H$ is $t^2 y / (1 + t^2 y^2) - \operatorname{artanh} y$. For $y > 0$, the quotient $\operatorname{artanh} y / y$ increases strictly from 1 to infinity, as its positive power series shows. The right side of (6.13) decreases strictly. For $t^2 \leq 1$ there is no positive crossing. For $t^2 > 1$ there is exactly one, and it is strict; symmetry gives the negative maximizer. These crossings are nondegenerate. For $\beta < \pi/2$, the second derivative in p at the center is $4(t^2 - 1) < 0$. Substituting it into the lattice Gaussian calculation gives $1/\sqrt{2\pi} \cos \beta$. The critical quartic case was already proved. $\square$

6.3 A uniform critical crossover and its first correction

The two sides of the angular transition fit into one uniform theorem. For $q > 0$ and real τ, η , set

$$\beta_N = \frac{\pi}{2} + \frac{\tau}{\sqrt{N}}, \quad \delta_N = \frac{\eta}{N^{3/4}}, \quad A_N = qe^{\delta_N/2}, \quad B_N = qe^{-\delta_N/2}. \quad (6.14)$$

Here $D_N(A, B; \beta)$ replaces $|m + in|$ in (6.1) by $|m\lambda_+ + n\lambda_-|$, with $\lambda_\pm = e^{\pm i\beta/2}$.

Theorem 6.4 (Uniform quartic profile). *Define*

$$\mathcal{P}(\tau, \eta) = \frac{\sqrt{2}}{\pi} \int_{\mathbb{R}} \exp\left(4\tau u^2 + \eta u - \frac{16}{3}u^4\right) du, \quad (6.15)$$

$$P_\tau(u) = \frac{\tau}{2} + 4\tau^2 u^2 - 16\tau u^4 + \frac{128}{15}u^6, \quad (6.16)$$

$$\mathcal{P}_1(\tau, \eta) = \frac{\sqrt{2}}{\pi} \int_{\mathbb{R}} P_\tau(u) \exp\left(4\tau u^2 + \eta u - \frac{16}{3}u^4\right) du. \quad (6.17)$$

Uniformly for (τ, η) in any compact subset of $\mathbb{R}^2$,

$$\frac{N^{5/4} D_N(A_N, B_N; \beta_N)}{[2eq \cos(\beta_N/2)]^N} = \mathcal{P}(\tau, \eta) + N^{-1/2} \mathcal{P}_1(\tau, \eta) + O(N^{-1}). \quad (6.18)$$

The factor q cancels exactly from the normalized expression.

Proof. Put $p = 1/2 + x$, $c_N = \cos(\beta_N/2)$, $t_N = \tan^2(\beta_N/2)$ and

$$\Xi_N(x) = H(1/2 + x) - \log 2 + \frac{1}{2} \log(1 + 4t_N x^2) + \delta_N x.$$

After removing the central exponential factor, uniform Stirling expansion on a fixed central interval gives each summand as

$$\frac{1 + O(N^{-1})}{2\pi N^2} \frac{\exp(N\Xi_N(x))}{c_N \sqrt{1 + 4t_N x^2} \sqrt{1/4 - x^2}}. \quad (6.19)$$

The entropy inequality $H(1/2 + x) \leq \log 2 - 2x^2 - 4x^4/3$, valid also by continuity at the endpoints, and $\log(1 + y) \leq y$ give

$$\Xi_N(x) \leq 2(t_N - 1)x^2 - \frac{4}{3}x^4 + \delta_N x.$$

Since $t_N - 1 = O(N^{-1/2})$, the coordinate $u = N^{1/4}x$ gives a uniform bound

$$N\Xi_N(uN^{-1/4}) \leq Cu^2 - \frac{4}{3}u^4 + E|u|. \quad (6.20)$$

Outside any fixed central interval the entropy upper bound for binomial coefficients makes the normalized sum exponentially small. This handles $p = 0, 1$ without an invalid endpoint Stirling approximation.

Taylor expansion gives $t_N = 1 + 2\tau N^{-1/2} + 2\tau^2 N^{-1} + O(N^{-3/2})$ and

$$\begin{aligned} N\Xi_N(uN^{-1/4}) &= 4\tau u^2 + \eta u - 16u^4/3 \\ &\quad + N^{-1/2}(4\tau^2 u^2 - 16\tau u^4 + 128u^6/15) \\ &\quad + O(N^{-1}(u^2 + u^4 + u^6 + u^8)). \end{aligned}$$

The smooth prefactor has a further cancellation:

$$\begin{aligned} \frac{1}{c_N \sqrt{1 + 4t_N x^2} \sqrt{1/4 - x^2}} &= \frac{2}{c_N} [(1 + 4t_N x^2)(1 - 4x^2)]^{-1/2} \\ &= 2\sqrt{2} \left[1 + \frac{\tau}{2\sqrt{N}} + O(N^{-1}(1 + u^2 + u^4)) \right]. \end{aligned}$$

There is no $u^2 N^{-1/2}$ contribution from this prefactor.

Choose the fixed central interval sufficiently small. Taylor's theorem for the real exponential, together with (6.20), then bounds the difference between the scaled summands and their first two displayed terms by

$$h_N N^{-1} Q(u) \exp(C_1 u^2 - c_1 u^4 + E|u|), \quad h_N = N^{-3/4}, \quad c_1 > 0,$$

where Q is a fixed polynomial. To see why the bound is uniform on the whole central interval, note that $N^{-1/2}u^6 \leq \epsilon^2 u^4$ when $|x| \leq \epsilon$; choosing ϵ small preserves negative quartic domination. The higher remainders are treated in the same way.

The u -grid has spacing h_N , independently of its parity-dependent shift. The functions $\exp(4\tau u^2 + \eta u - 16u^4/3)$ and its product with P_τ , including their first two derivatives, have uniformly bounded integrals of absolute value on compact parameter sets. The midpoint-cell estimate

$$\left| h \sum_{k \in \mathbb{Z}} f(a + kh) - \int_{\mathbb{R}} f(u) du \right| \leq Ch^2 \|f''\|_{L^1}$$

follows by integrating Taylor's formula on each centered cell. It makes the lattice error $O(N^{-3/2})$. The integrable remainder above sums to $O(N^{-1})$. Multiplying the central prefactor by the mesh factor proves (6.18). $\square$

At the critical point $\mathcal{P}(0,0) = C_4$. The profile extends to an entire function of its two complex parameters, is even in η , and satisfies

$$\partial_\tau \mathcal{P} = 4\partial_\eta^2 \mathcal{P}, \quad \frac{64}{3} \partial_\eta^3 \mathcal{P} - 8\tau \partial_\eta \mathcal{P} - \eta \mathcal{P} = 0. \quad (6.21)$$

Both identities follow by differentiation under the quartically dominated integral and integration by parts. They provide a convenient way to generate its moment corrections. The theorem identifies $N^{-1/2}$ as the angle scale and $N^{-3/4}$ as the amplitude-imbalance scale; it is more informative than separate limiting exponents.

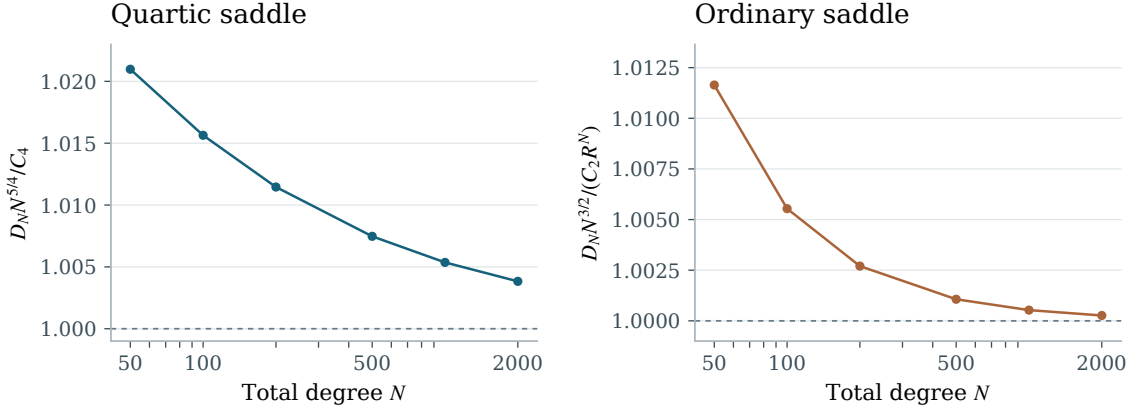


Figure 2: Checks of the quartic and Gaussian degree laws. The balanced case is normalized by $C_4 N^{-5/4}$ at its convergence boundary. The unbalanced case uses its own growth factor and Gaussian prefactor. The proof rests on the localized lattice-saddle argument, not the finite numerical comparisons.

6.4 A cubic critical point reached by the selected branch

Take

$$a = b = \frac{1-i}{2}, \quad F(z) = z + ae^{-z} + be^{-iz}, \quad w_0 = 1-i. \quad (6.22)$$

Direct calculation gives

$$F(0) = w_0, \quad F'(0) = F''(0) = 0, \quad F'''(0) = i. \quad (6.23)$$

Both effective moduli at w_0 are $1/(e\sqrt{2})$, so this critical value lies exactly on the symmetric absolute-convergence boundary.

To identify the branch, scale both amplitudes by $t \in [0, 1]$ and set $z = (1-i)v$. The equation $F_t(z) = w_0$ becomes

$$v + te^{-v} \cos v = 1, \quad t(v) = \frac{(1-v)e^v}{\cos v}, \quad 0 \leq v \leq 1.$$

It maps $v = 1$ to $t = 0$ and $v = 0$ to $t = 1$ and is strictly decreasing between them, because $(\log t)' = \tan v - v/(1-v) < 0$. For the inequality, $v \cos v - (1-v) \sin v$ vanishes at zero and has positive derivative $(1-v) \sin v + v \cos v$ on $(0, 1)$. Along this path, $F'_t((1-i)v) = v - (1-v) \tan v > 0$ for $0 < v < 1$, so the inverse branch stays simple before the endpoint. Thus the branch selected at zero amplitude reaches $z = 0$ at $t = 1$. Moreover $t(v) = 1 - v^3/3 + O(v^4)$.

For the fixed function F , its three local inverse branches have

$$z = \left(\frac{6(w - w_0)}{i} \right)^{1/3} + O((w - w_0)^{2/3}). \quad (6.24)$$

The exponent $1/3$ is an inverse ramification exponent. The exponent $5/4$ concerns sums of absolute values of bivariate coefficients at one degree. Complex phase cancellations distinguish the quantities; one exponent should not be read off from the other.

7 A sharp Borel majorant and inverse-domain certificate

The inverse Borel formula and the exponential bound $Me^{(\tau+2\sqrt{M})|\zeta|}$ already appear in the repository's *Negative-Ray Summation of Exponential Feedback*, Lemma `lem:fineinverse` [5].

The refinement here is the exact Bessel majorant, its pointwise optimality, and the matching source and target half-plane bounds. The next section applies them uniformly to sine-period families.

Let K be holomorphic near zero and on a star-shaped open sector of the Borel plane. Assume, on the rays being used, that

$$|K(\zeta)| \leq M e^{\tau|\zeta|}, \quad M > 0, \quad \tau \geq 0. \quad (7.1)$$

Let $\tilde{h} = \mathcal{B}^{-1}K$ and let $\tilde{G} = z + \tilde{v}$ be the formal inverse of $z + \varepsilon\tilde{h}(z)$.

Theorem 7.1 (Optimal Bessel majorant). *Set $a = |\varepsilon|M$. The inverse Borel transform is*

$$\hat{v}(\zeta) = - \sum_{n \geq 1} \frac{\varepsilon^n}{n!} \zeta^{n-1} K^{*n}(\zeta). \quad (7.2)$$

It converges normally on the same star sector and satisfies, for $r = |\zeta|$,

$$|\hat{v}(\zeta)| \leq a e^{\tau r} \frac{I_1(2\sqrt{a}r)}{\sqrt{a}r} \leq a e^{(\tau+2\sqrt{a})r}. \quad (7.3)$$

The quotient is defined as 1 at $ar^2 = 0$. Both the pointwise majorant and its universal exponential-type bound are sharp.

Proof. Formal Lagrange inversion gives $\tilde{v} = \sum_{n \geq 1} (-\varepsilon)^n D^{n-1}(\tilde{h}^n)/n!$. Its n th term has order at least $z^{-(2n-1)}$, so substitution of ε is formally legitimate coefficient by coefficient. The Borel rules in (2.1) give (7.2).

The n -fold straight-segment convolution is an integral over a simplex of volume $r^{n-1}/(n-1)!$. The lengths of its arguments sum to r . Consequently

$$|K^{*n}(\zeta)| \leq M^n e^{\tau r} \frac{r^{n-1}}{(n-1)!}. \quad (7.4)$$

Summing gives

$$|\hat{v}(\zeta)| \leq e^{\tau r} \sum_{n \geq 1} \frac{a^n r^{2n-2}}{n!(n-1)!} = a e^{\tau r} \frac{I_1(2\sqrt{a}r)}{\sqrt{a}r},$$

where the equality can also be taken as the defining power series of this Bessel quotient. Its normal convergence justifies the analytic Borel formula. For $t \geq 0$,

$$\frac{I_1(2t)}{t} = \sum_{k \geq 0} \frac{t^{2k}}{k!(k+1)!} \leq \cosh(2t) \leq e^{2t};$$

the first inequality uses $(2k)!/(k!(k+1)!) \leq 4^k$.

For sharpness choose $K(\zeta) = M e^{\tau\zeta}$ and real $\varepsilon > 0$. Then $K^{*n} = M^n e^{\tau\zeta} \zeta^{n-1}/(n-1)!$. All terms of (7.2) have the same sign on the positive ray, so equality holds in the first bound. The integral identity

$$\frac{I_1(2t)}{t} = \frac{2}{\pi} \int_{-1}^1 e^{2tu} \sqrt{1-u^2} du$$

follows by expanding the exponential and evaluating beta integrals. It shows that its logarithmic growth rate is 2: the upper bound was proved, and restriction to $u \in [1-\delta, 1-\delta/2]$ gives every lower exponential rate $2(1-\delta)$. Thus the attained rate in (7.3) is $\tau + 2\sqrt{a}$. $\square$

Theorem 7.2 (Sharp half-plane inversion certificate). *Fix a Borel direction θ satisfying (7.1), and set $h_\theta = \mathcal{L}_\theta K$, $F_\theta = z + \varepsilon h_\theta$. Then F_θ is univalent on*

$$\operatorname{Re}(e^{i\theta} z) > \tau + \sqrt{a}. \quad (7.5)$$

Its image contains the target half-plane

$$\operatorname{Re}(e^{i\theta} w) > \tau + 2\sqrt{a}. \quad (7.6)$$

There the inverse is $G_\theta(w) = w + \mathcal{L}_\theta \hat{v}(w)$ and, with $q = \operatorname{Re}(e^{i\theta} w) - \tau$,

$$|G_\theta(w) - w| \leq \frac{q - \sqrt{q^2 - 4a}}{2}. \quad (7.7)$$

The source and target thresholds are optimal as universal constants under (7.1).

Proof. Direct Laplace estimates give

$$|\varepsilon h_\theta(z)| \leq \frac{a}{q_z}, \quad |\varepsilon h'_\theta(z)| \leq \frac{a}{q_z^2}, \quad q_z = \operatorname{Re}(e^{i\theta} z) - \tau > 0.$$

On the convex half-plane (7.5), integrate the derivative of the perturbation along each line segment. Its modulus is strictly below one on that segment, proving injectivity.

If $q > 2\sqrt{a}$, use the disk centered at w of radius $q/2$. On it the perturbation has norm at most $2a/q < q/2$. Rouché gives a unique root with displacement $d < q/2$, and the root equation gives $d \leq a/(q - d)$. Since $d(q - d)$ increases on $[0, q/2]$, this implies (7.7). Its image lies in the source half-plane, because $q - d \geq (q + \sqrt{q^2 - 4a})/2 > \sqrt{a}$.

Termwise Laplace integration of (7.2) is justified by the Bessel majorant. Its positive integral is exactly

$$\sum_{n \geq 1} \frac{(2n-2)!}{n!(n-1)!} \frac{a^n}{q^{2n-1}} = \frac{q - \sqrt{q^2 - 4a}}{2}. \quad (7.8)$$

The coefficients are Catalan numbers. The summed function agrees with the analytic Lagrange inverse for q sufficiently large and hence, by analyticity and the preceding range inclusion, throughout (7.6).

For $K = Me^{\tau\zeta}$, $\theta = 0$, $\varepsilon > 0$, one has $F(z) = z + a/(z - \tau)$ and

$$G(w) = \frac{w + \tau + \sqrt{(w - \tau)^2 - 4a}}{2}, \quad (7.9)$$

with the square root selected at infinity. The positive critical source point is $\tau + \sqrt{a}$ and its value is $\tau + 2\sqrt{a}$. Equality also holds in (7.7) on the positive real target ray. Neither universal threshold can be lowered. $\square$

The sign distinction is useful: $\hat{v}$ is negative on the positive ray in the sharp model. The positive expression in (7.8) is the majorant for $|G - w|$, not the signed displacement $G - w$.

8 A sine-kernel family: uniform bounds and filtered continuation

The next application separates two questions that should not be conflated: uniform estimates on straight nonreal summation rays, and continuation along more general paths that may reach other sheets. The first follows from an elementary estimate; the second uses a published nonlinear continuation theorem.

8.1 Uniform inversion away from real Borel directions

For $\alpha > 0$ consider the meromorphic kernel

$$K_\alpha(\zeta) = \frac{\zeta^2}{\sin(\pi\zeta)\sin(\pi\alpha\zeta)}. \quad (8.1)$$

The origin is removable, with $K_\alpha(0) = 1/(\pi^2\alpha)$. Its possible poles belong to

$$\Sigma_\alpha = (\mathbb{Z} \cup \alpha^{-1}\mathbb{Z}) \setminus \{0\}.$$

At rational α , nonzero poles from the two lattices can coincide. At irrational α , distinct poles can approach one another arbitrarily closely at large distance. Splitting the kernel into individual residues is therefore poorly suited to estimates uniform in α .

Theorem 8.1 (Period-uniform sectorial inverse bounds). *Fix $0 < \alpha_0 \leq \alpha_1 < \infty$ and $0 < \eta \leq \pi/2$. For every $\alpha \in [\alpha_0, \alpha_1]$ and every ray angle θ with $|\sin \theta| \geq \sin \eta$,*

$$|K_\alpha(re^{i\theta})| \leq M_\eta := \frac{1}{\pi^2\alpha_0 \sin^2 \eta}, \quad r \geq 0. \quad (8.2)$$

Let $h_{\alpha,\theta} = \mathcal{L}_\theta K_\alpha$ and $a_\eta = |\varepsilon| M_\eta$. The map $F_{\alpha,\theta} = z + \varepsilon h_{\alpha,\theta}$ is univalent on

$$\operatorname{Re}(e^{i\theta} z) > \sqrt{a_\eta},$$

and its image contains

$$\operatorname{Re}(e^{i\theta} w) > 2\sqrt{a_\eta}.$$

On the latter half-plane, its inverse satisfies

$$|G_{\alpha,\theta}(w) - w| \leq \frac{q - \sqrt{q^2 - 4a_\eta}}{2}, \quad q = \operatorname{Re}(e^{i\theta} w),$$

and the inverse Borel transform has the uniform bound

$$|\widehat{v}_\alpha(re^{i\theta})| \leq a_\eta \frac{I_1(2\sqrt{a_\eta} r)}{\sqrt{a_\eta} r} \leq a_\eta e^{2\sqrt{a_\eta} r}.$$

These statements hold uniformly through rational values of α .

Proof. The elementary identity

$$|\sin(x + iy)|^2 = \sin^2 x + \sinh^2 y$$

gives $|\sin(x + iy)| \geq \sinh |y| \geq |y|$. Applying it to the two denominator factors yields, for $r > 0$,

$$|K_\alpha(re^{i\theta})| \leq \frac{r^2}{\pi r |\sin \theta| \pi \alpha r |\sin \theta|} \leq M_\eta.$$

Continuity gives the same bound at zero. The preceding Borel and half-plane theorems now apply with $\tau = 0$.

For completeness, continuity in α follows directly from the unsplit kernel. On every compact part of an off-axis ray, the removable value at zero and the nonvanishing denominators give continuous dependence on α . The uniform bound controls the Laplace tail on every strictly smaller target half-plane. The Bessel majorant also permits termwise passage to the limit in the inverse Borel series. Coinciding real poles create no exceptional case. $\square$

This theorem is a uniformity statement, not an assertion that M_η is optimal for this particular family. Optimality in [theorem 7.2](#) concerns the full class of kernels with a given bound. Also, M_η diverges as $\eta \downarrow 0$. The argument gives neither a uniform approach to a real Stokes direction nor uniform bounds on arbitrary continuation sheets.

As a normalization check, the formal Laplace series begins

$$h_\alpha(z) = \frac{1}{\pi^2 \alpha z} + \frac{1 + \alpha^2}{3\alpha z^3} + O(z^{-5}).$$

These are coefficients of the asymptotic formal series; the actual function is specified by the chosen summation direction.

8.2 The continuation-cost set

Suppose now that α is irrational, and write $\ell_1 = 1$, $\ell_2 = \alpha^{-1}$. Define a family of finite sets by

$$\mathcal{S}_L = \left\{ \omega_1 + \cdots + \omega_n : \begin{array}{l} n \geq 1, \quad \omega_j \in \Sigma_\alpha, \\ |\omega_1| + \cdots + |\omega_n| \leq L \end{array} \right\}. \quad (8.3)$$

The budget records the sum of the costs of the primitive pieces. It is not the modulus of their final sum.

Proposition 8.2 (Explicit saturated cost). *For irrational α ,*

$$\begin{aligned} \mathcal{S}_L = & \{ m\ell_1 + n\ell_2 : (m, n) \in \mathbb{Z}^2 \setminus \{(0, 0)\}, \\ & |m|\ell_1 + |n|\ell_2 \leq L \} \\ & \cup \begin{cases} \{0\}, & L \geq 2 \min(\ell_1, \ell_2), \\ \emptyset, & L < 2 \min(\ell_1, \ell_2). \end{cases} \end{aligned} \quad (8.4)$$

Every $\mathcal{S}_L$ is finite, whereas $\bigcup_{L \geq 0} \mathcal{S}_L = \ell_1 \mathbb{Z} + \ell_2 \mathbb{Z}$ is dense in $\mathbb{R}$. The family is empty for sufficiently small L , is increasing and right-continuous in L , and satisfies

$$\mathcal{S}_{L_1} + \mathcal{S}_{L_2} \subseteq \mathcal{S}_{L_1 + L_2}.$$

Proof. Irrationality makes the integer representation $m\ell_1 + n\ell_2$ unique. In any primitive word representing this point, separate the two lattices and apply the triangle inequality to each integer sum. Its cost is at least $|m|\ell_1 + |n|\ell_2$. If $(m, n) \neq (0, 0)$, this lower bound is attained by using the nonzero terms $m\ell_1$ and $n\ell_2$, omitting a zero term. A nonempty word representing zero must have zero total integer coefficient in each lattice. It must therefore contain a cancellation within at least one lattice. Its cost is at least $2 \min(\ell_1, \ell_2)$, attained by the shortest opposite pair.

The explicit integer bounds prove finiteness at each budget and right-continuity. Concatenating words proves the additive inclusion. To see density, irrationality and the pigeonhole principle produce nonzero elements of $\ell_1 \mathbb{Z} + \ell_2 \mathbb{Z}$ of arbitrarily small modulus. Their integer multiples approximate any real number to within half that modulus. $\square$

The delayed zero in (8.4) is essential. The original Borel germ is regular at zero. The empty word must not place zero in the forbidden set at cost zero. A possible singularity over zero reached after a nontrivial continuation history is a different object.

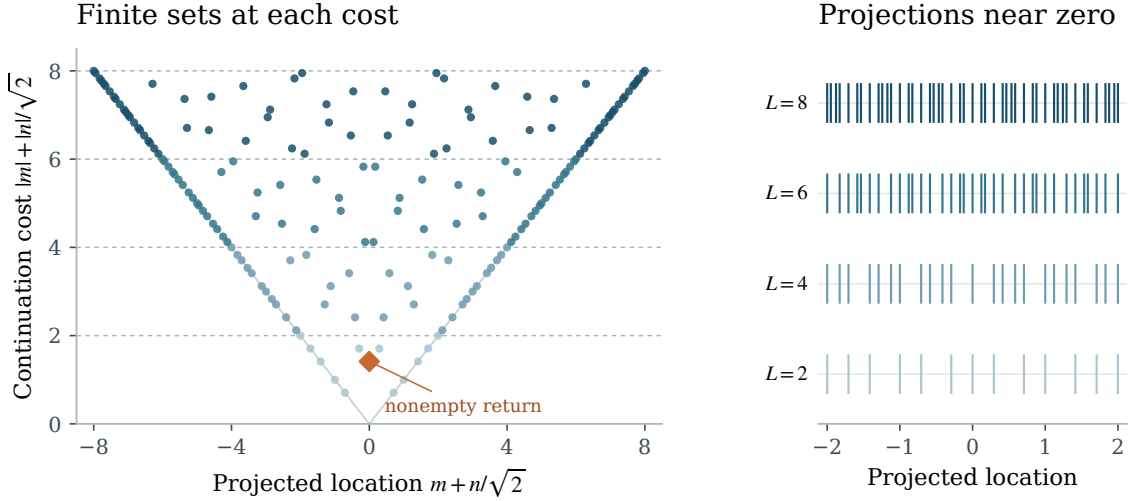


Figure 3: Potential Borel locations for $\alpha = \sqrt{2}$, filtered by the cost $|m| + |n|/\sqrt{2}$. The projection is globally dense, while every fixed budget contains finitely many locations. Zero appears only after the nonempty return cost $\sqrt{2}$. The figure displays possible locations; it does not assert that each is singular.

8.3 An application of filtered implicit-function theory

A discrete filtered set is, in the terminology of Kamimoto and Sauzin [13, Definitions 2.1–2.8], a family of finite subsets of $\mathbb{C}$ that increases with a length budget and is empty at sufficiently small budget, with the relevant upper closure used in defining admissible paths. For the right-continuous family above, a piecewise smooth path γ starting at zero is allowed when

$$\gamma(t) \notin \mathcal{S}_{\text{length}(\gamma|_{[0,t]})} \quad \text{for all } t.$$

The condition is imposed on the pair consisting of accumulated length and current position, not merely on the image of the path.

Theorem 8.3 (Filtered continuation of the inverse). *Let h_α be the formal inverse-power series with $\mathcal{B}h_\alpha = K_\alpha$, and let $G_\alpha(w) = w + v_\alpha(w)$ be the formal compositional inverse of $z + \varepsilon h_\alpha(z)$. For irrational $\alpha > 0$, $\mathcal{B}v_\alpha$ admits analytic continuation along every $\mathcal{S}$ -allowed path, with $\mathcal{S}$ given by (8.3)–(8.4). For rational $\alpha > 0$, the same conclusion holds with the word-cost definition (8.3), with coincident locations identified.*

Proof. This is an application of the Kamimoto–Sauzin implicit-function theorem [13, Theorem 5.3]; we verify its relevant inputs rather than reprove its continuation estimates.

Start with the filtered set $\Omega_L = \{\omega \in \Sigma_\alpha : |\omega| \leq L\}$. It is finite at fixed L . The meromorphic function K_α is holomorphic along every Ω -allowed path, because a path that reaches a pole ω has length at least $|\omega|$. Multiplication of K_α by a polynomial creates no new singular locations. Thus every coefficient in

$$\mathcal{F}(z, u) = u + \varepsilon \sum_{k \geq 0} \frac{h_\alpha^{(k)}(z)}{k!} u^k \quad (8.5)$$

has the required filtered continuation. For the compact sets $K_\Omega^{\delta, L}$ of [13, Notation 3.15], on the continuation surface with projection p_Ω , put

$$M_{\delta, L} = \sup_{\xi \in K_\Omega^{\delta, L}} |K_\alpha(p_\Omega(\xi))| < \infty.$$

Every such ξ is reached by a path of length at most L , so $|p_\Omega(\xi)| \leq L$. Writing $\mathcal{F}(z, u) = \sum_{k \geq 0} \mathcal{F}_k(z) u^k$, the seminorms of [13, Definition 3.16] satisfy

$$\|\mathcal{F}_k\|_\Omega^{\delta, L} \leq \mathbf{1}_{\{k=1\}} + |\varepsilon| M_{\delta, L} \frac{L^k}{k!} \leq C_{\delta, L}^{k+1} \quad (k \geq 0)$$

for a suitable $C_{\delta, L} \geq 1$. This verifies precisely the coefficient-growth condition in [13, Definition 5.1], including the scalar u term. The constants here can depend on the fixed α and ε ; no uniformity on later sheets is inferred.

The constant term at $z = \infty$ of $\mathcal{F}$ is u , so its value at $u = 0$ is zero and its u derivative there is one. The unique formal implicit solution therefore exists, and its Borel transform is continuable with respect to the saturation $\Omega^{*\infty}$. The filtered convolution operation adds the primitive costs; its saturation is exactly the nonempty-word set (8.3). This proves the claim. $\square$

No step of this theorem proves that every element of $\mathcal{S}_L$ is an actual singularity. The theorem gives a controlled continuation surface and an upper bound on possible obstructions. The same distinction appears in Kamimoto and Sauzin's discussion of dense projected singular sets [13, Remark 2.17]. It would be incorrect to infer a natural boundary from density of the unfiltered projection alone.

The two results in this section complement each other. Theorem 8.1 supplies explicit constants, uniform in α , on nonreal straight rays. Theorem 8.3 supplies a broader qualitative continuation statement for a fixed parameter. Uniform bounds on later sheets, especially while real poles coalesce, remain an additional problem.

9 Finite Stokes transport, logarithmic cores, and the limits of generality

9.1 The exact transition identity

The inverse of a sectorial jump is nonlinear. This fact is already developed in the repository's nonlinear Stokes report [3] and in the general resurgent calculus [11]. It fits naturally with the exact-core theorem.

Let F_- and F_+ be two holomorphic realizations with compatible inverse branches G_- and G_+ , and set $J = F_+ - F_-$ on their common source domain. On a target domain where the compositions are defined,

$$T = F_+ \circ G_- = \text{id} + J \circ G_-, \quad G_+ = G_- \circ T^{-1}. \quad (9.1)$$

If $J \circ G_-$ is small in the disk norm of theorem 4.1, then

$$G_+ - G_- = \sum_{n \geq 1} \frac{(-1)^n}{n!} \text{D}^{n-1} [G'_- (J \circ G_-)^n]. \quad (9.2)$$

In particular, the first variation is $-G'_- J \circ G_-$, and the higher variations generate all mixed jump actions.

The proof of (9.1) is composition: $F_+ \circ G_- = T$ implies $F_+ \circ G_- \circ T^{-1} = \text{id}$ on the chosen branch. Equation (9.2) is then theorem 4.1 with exact core G_- . If several realizations are indexed by α, β, γ , their target transition maps satisfy

$$T_{\gamma\beta} \circ T_{\beta\alpha} = T_{\gamma\alpha}, \quad T_{\beta\alpha} = F_\beta \circ G_\alpha,$$

whenever the same intermediate inverse branch is used and all compositions are defined. This is an analytic transition identity. The dense sorting walls in [section 5](#) are not, by themselves, such jumps.

9.2 A logarithmic core and complex powers

Consider, on a fixed logarithm branch,

$$F(z) = z + \rho \log z + be^{-\lambda z}. \quad (9.3)$$

Let G_0 be a chosen holomorphic inverse of $z + \rho \log z$, away from points where $G_0 + \rho = 0$. Implicit differentiation gives

$$G'_0 = \frac{G_0}{G_0 + \rho}, \quad G''_0 = \frac{\rho G_0}{(G_0 + \rho)^3}, \quad G'''_0 = \frac{\rho G_0(\rho - 2G_0)}{(G_0 + \rho)^5}. \quad (9.4)$$

With $H = be^{-\lambda G_0}$, the exact-core expansion begins

$$\begin{aligned} G &= G_0 - bG'_0 e^{-\lambda G_0} \\ &\quad + \frac{b^2}{2} (G''_0 - 2\lambda(G'_0)^2) e^{-2\lambda G_0} \\ &\quad - \frac{b^3}{6} (G'''_0 - 9\lambda G'_0 G''_0 + 9\lambda^2(G'_0)^3) e^{-3\lambda G_0} + \dots \end{aligned} \quad (9.5)$$

This is convergent whenever the disk smallness hypothesis holds, and otherwise is a formal amplitude expansion under the appropriate formal interpretation.

On an admissible asymptotic branch,

$$G_0(w) = w - \rho \log w + O(\log w/w).$$

Consequently,

$$e^{-\lambda G_0(w)} = e^{-\lambda w} w^{\rho\lambda} (1 + O(\log w/w)), \quad (9.6)$$

with the complex power defined by that same logarithm branch. The exponential action remains λ , but the coefficient alphabet acquires complex powers and logarithms. This example explains why preservation of the additive action semigroup is a weaker assertion than preservation of a prescribed full transseries alphabet.

The formulas also show where the regular core chart fails: $G_0 + \rho = 0$ makes the displayed derivatives singular. A Puiseux or another critical inverse chart must replace regular Lagrange inversion there. The repository already treats several real positive-action critical regimes [\[6, 7\]](#); [section 6](#) exhibits a genuinely complex cubic example.

9.3 What general transseries require beyond this article

The weighted theorem applies to arbitrary holomorphic phases Φ_j , including nonlinear phases, as soon as the stated disk norms and coefficient controls are proved. For example, perturbations built from e^{-z^2} or e^{-e^z} can be handled locally on suitable domains. The finite linear-action cone theorem does not automatically establish those domain estimates: a small phase on one ray can grow rapidly in a disk or in a neighboring direction.

Likewise, a formal inverse in a degree completion does not by itself choose an analytic realization. Distinct holomorphic functions may have the same power-log asymptotic expansion and differ by an exponentially small term. Their inverses need not agree. A summation

prescription, an exact analytic core, or another branch normalization is therefore part of a meaningful analytic inversion statement.

Finally, local or sectorial inverse closure does not imply one single-valued inverse on the entire complex plane. Already $F(z) = z + e^{-z}$ has critical points $2\pi ik$ and critical values $1 + 2\pi ik$, $k \in \mathbb{Z}$. Any global continuation theory has to keep track of inverse sheets and their monodromy. Unrestricted complex transseries encompass ordered algebraic, sectorial analytic, multisummable, resurgent, and higher-level questions with different hypotheses. The results here give compatible pieces of that theory rather than identify these categories.

10 Further research questions and a proposed programme

The questions below distinguish extensions that have a clear finite model from problems requiring a broader analytic category. They are proposed problems, not claims of established open status throughout the literature. Their relationship to the inspected repository is described in [appendix A](#).

10.1 Exact convergence and critical asymptotics

Research question 10.1 (More than two complex actions). For $F(z) = z + \sum_{j=1}^s a_j e^{-\lambda_j z}$, determine the full absolute-convergence domain of its labelled inverse expansion. The natural variational quantity on the simplex is

$$\Psi(\mathbf{p}) = 1 + \log \left| \sum_j p_j \lambda_j \right| - \sum_j p_j \log p_j + \sum_j p_j \log A_j, \quad A_j = |a_j e^{-\lambda_j w}|.$$

Which geometric hypotheses make the maximizer unique? Which degeneracies are possible, and which boundary degree laws are summable? A complete result should treat faces of the simplex, zeros of the action barycenter, and exact action collisions.

The two-action proof suggests a useful first target: three actions in a strict cone, with an interior maximizer and a Hessian that is negative definite except on one controlled degeneracy locus. Ordinary interior saddles are expected to retain the degree exponent $-3/2$ after the dimensional factors cancel. A proof should specify the hypotheses under which that cancellation remains uniform.

Research question 10.2 (Criticality of the majorant versus the actual inverse). Classify the relation between degeneracies of the absolute degree sum and ramification of the actual complex inverse. The orthogonal example has a quartic saddle in the absolute sum and a cubic critical point for a suitable phase choice. When does such a correspondence hold, and when does oscillatory cancellation separate the two phenomena?

A useful answer would distinguish three boundaries: the domain of absolute convergence of the labelled multivariate series, the domain of its analytic continuation, and the critical-value set of the selected inverse sheet. These sets need not coincide. Generalized Lambert inverses and multivariate density inversion provide related inversion settings [16, 17, 18], but the additive complex-exponential geometry here requires its own analysis.

Research question 10.3 (All-order uniform crossover). Extend [theorem 6.4](#) to an all-order expansion with explicit recursively computable polynomial corrections and a uniform remainder. Determine matching expansions when $|\tau| + |\eta|$ grows with N , so that the quartic regime joins the one-saddle and two-saddle Gaussian regimes.

The first correction already shows the mechanism: Taylor expansion of the exponent and amplitude produces polynomial moments of $\mathcal{P}$. The differential equations in (6.21) can reduce those moments to a finite set of derivatives. The remaining work is uniform control in enlarging parameter ranges and near competing saddles.

10.2 Angular support and higher transseries levels

Research question 10.4 (Moving walls with nonconstant amplitudes). Construct finite angular atlases for terms $w^\beta (\log w)^k e^{-\Lambda w}$, with explicit error bounds uniform near crossings. For $w = Re^{i\theta}$, power amplitudes can shift a simple exponential-rate crossing by $O(\log R/R)$. What is the correct finite description when several crossings coalesce or when the first angular derivative vanishes?

This problem should retain both exponential accuracy and algebraic accuracy. Sorting only the actions is sufficient for the former away from degenerate ties, but does not order full terms uniformly. A useful deliverable is an algorithm that returns an approximation with a certified remainder on each angular cell, including a local transition formula where the cell boundaries move with R .

Research question 10.5 (Nonlinear phases and several levels of smallness). Find a checkable replacement for the strict linear-action cone criterion for phases $\Phi_j(w)$ from logarithmic-exponential or higher transseries classes. The criterion should control differentiation, composition with an inverse core, and uniform disk smallness on complex domains.

For polynomial phases, a first step is to stratify directions by the leading nonzero real part and then retain lower-degree phases on the resulting boundary directions. For nested exponentials, one needs a hierarchy of domains and derivative bounds. An action-degree filtration alone does not provide this information.

Research question 10.6 (Countable actions and cone boundaries). Replace finite generation by a weighted local-finiteness condition that is both necessary and sufficient for finite coefficient fibers and useful analytic inverse tails. Include actions approaching a cone boundary, and determine when oscillatory boundary terms can be retained without destroying composition.

The hypotheses should separate the number of labels, the size of their coefficients, and the cost of their actions. The weighted theorem tolerates countably many labels analytically, while a finite action cutoff needs an additional proper weight. Reconciling these requirements with cancellation and accumulation would connect the present work to the repository's action-measure and Hahn-support developments.

10.3 Resurgent continuation and resonance clusters

Research question 10.7 (Constructing complex resonance blocks). Prove cluster estimates for meromorphic Borel kernels with genuinely complex pole rays, using a common family of Laplace domains. The target is a postgrouping norm bound of the form required in [theorem 4.1](#), uniform as poles within a cluster collide.

This is the missing analytic construction behind the complex-cone question in [4]. The cone theorem controls action addition after suitable blocks and tails have been obtained; it does not manufacture those blocks. A successful proof should estimate complete clusters and avoid constants that depend on individual inverse pole separations.

Research question 10.8 (Actual inverse Borel singularities). For (8.1), determine which points in the cost set $\mathcal{S}_L$ are actual singularities of $\mathcal{B}v_\alpha$ on specified sheets. Compute their local germs, and determine whether the minimal word cost equals the minimal continuation length needed to encounter each singularity.

Even the first nonprimitive points would be informative. The convolution formula gives a route to local singularity calculations, but the infinite inverse series can create cancellations. A theorem should identify the sheet and local continuation path, rather than list projected locations alone.

Research question 10.9 (Uniform continuation through period coalescence). Extend the off-axis estimates of theorem 8.1 to compact families of allowed continuation paths on later sheets, uniformly as α passes through rational values. What renormalized bounds remain valid when the paths approach the real singular directions?

The explicit kernel shows that first-sheet off-axis uniformity is compatible with pole coalescence. On other sheets, the interaction of convolution singularities, path geometry, and resonance blocks must be controlled together. The blow-up of the elementary bound as $\eta \downarrow 0$ identifies where a different method is needed.

10.4 A compatible global framework and effective proofs

Research question 10.10 (Several compatible filtrations). Construct an inversion calculus retaining simultaneously amplitude degree, complex-action decay, continuation length, and Gevrey level. Under what hypotheses do completion, summation, composition, and inversion commute?

The examples in this article explain why no single scalar order captures all four roles. A satisfactory framework should provide comparison maps between finite quotients and analytic realizations, with precise statements of which maps preserve which operations. The transition identity (9.1) supplies one compatibility condition for the resulting sectorial charts.

Research question 10.11 (Critical charts and inverse monodromy). Develop a complex multi-action theory of critical inverse charts, including folds, cubic points, and higher ramification. Determine how their local fractional-power expansions, finite angular atlases, and resurgent transition maps glue along paths in the target plane.

The regular weighted theorem should be used where the exact core is nonsingular. Critical charts should take over at its failures, with overlap estimates proving that both representations select the same branch. The explicit cubic example provides a compact test case for such a construction.

Research question 10.12 (Certified computation and formalization). Give effective algorithms for finite inverse atlases and coefficient quotients, with bit-complexity bounds for algebraic or otherwise computably specified actions. Formalize a sequence of foundational lemmas and then the analytic inverse certificates in a proof assistant.

An efficient order is: exact mixed coefficients and finite fibers; the cone separation and properness equivalence; finite wall arrangements; Rouché and Cauchy-based remainder certificates; and finally the filtered continuation imports. For arbitrary real parameters, an oracle for action equality and sign comparison is an additional assumption, not an algorithm. The verification scripts accompanying this report are executable checks, not proof-assistant formalizations.

10.5 Three near-term projects with definite completion criteria

1. **Three-action pressure theorem.** Prove a precise variational convergence criterion for three noncollinear actions in a strict cone, including one controlled degeneracy and its boundary exponent. Completion means a theorem with a closed boundary criterion, not only a sufficient convergence radius.
2. **First nonprimitive Borel germs.** For the sine kernel at a fixed irrational period, compute the inverse germs at a specified pair of first convolution locations on named sheets and compare their onset costs with (8.4). Completion means local expansions with proven nonvanishing or cancellation statements.
3. **Uniform moving-wall approximation.** Add complex powers and logarithms to the two-action inverse and prove an approximation uniform across one simple sorting wall, including its displacement and a numerical certificate. Completion means an error estimate valid on both sides and in the crossing region.

11 Reproducible verification and numerical illustrations

The proofs above establish the assertions. The accompanying code checks coefficient identities, concrete truncation errors, and asymptotic constants independently of the typeset derivations. Exact arithmetic is used where available. Numerical measurements use 100 or 110 decimal digits and are floating-point consistency checks, not interval enclosures.

11.1 Exact checks

The script `verification/verification.py` works over exact Gaussian rationals to verify all 65 nonconstant coefficients through total degree 10 in

$$\delta + xe^{-\delta} + ye^{-i\delta} = 0.$$

It compares the explicit mixed formula both with direct substitution and with a separately constructed formal fixed-point iteration. It also enumerates the actual pairwise action differences in every cumulative truncation $N = 1, \dots, 12$ and checks the totient wall formula. Thus it tests the wall definition as well as the closed count.

The first 25 Bessel–Catalan coefficient identities and 24 quadratic recurrences are checked with rational arithmetic. For the cost illustration at $\alpha = \sqrt{2}$, inclusion in the displayed integer budgets is decided by an integer-square comparison, avoiding a floating-point decision at the boundary. The counts for $L = 2, 4, 6, 8$ are 13, 47, 105, 183, respectively, including zero at its delayed return cost.

11.2 Inverse truncation errors

Take $a = 1/3$, $b = 1/4$ in (1.1). The disk radius $r = 1$ gives the sufficient norm parameter

$$\kappa = e \left(|a|e^{-\operatorname{Re} w} + |b|e^{\operatorname{Im} w} \right) < 1.$$

The root is computed independently at 110-digit precision, with the equation residual required to be below 10^{-100} . Every measured error in table 2 lies below the analytic bound (4.6).

The Bessel kernel is also integrated numerically at $a = 1/4$, $q = 2$. Its positive Laplace integral agrees with $(q - \sqrt{q^2 - 4a})/2$ to absolute error below 10^{-100} . The sign convention is checked separately: this is $q - G(q)$ in the extremal model, rather than $G(q) - q$.

w	N	Measured error	Proven tail bound
$4 - 3i$	1	2.9378×10^{-4}	1.3391×10^{-3}
	2	6.289×10^{-6}	4.502×10^{-5}
	4	3.9584×10^{-9}	6.8695×10^{-8}
	8	3.4817×10^{-15}	2.4683×10^{-13}
	12	4.7494×10^{-21}	1.1052×10^{-18}
$6 - 4i$	1	1.6345×10^{-5}	1.0955×10^{-4}
	2	1.0713×10^{-7}	1.073×10^{-6}
	4	6.9236×10^{-12}	1.3899×10^{-10}
	8	5.3761×10^{-20}	3.5984×10^{-18}
	12	5.7518×10^{-28}	1.161×10^{-25}

Table 2: Inverse truncation errors and the proven disk bound, with $C = r = 1$. Displayed measured errors are numerical values, not rigorous intervals.

11.3 Quartic and crossover checks

The second script, `verification/quartic_saddle_checks.py`, evaluates the degree sums by a stable log-sum-exp calculation. For the symmetric boundary $A = B = 1/(e\sqrt{2})$, it compares $D_N N^{5/4}$ with

$$C_4 = 0.53699040102233754294 \dots$$

For $A = 0.2$, $B = 0.1$, it checks the ordinary saddle with

$$p_* = 0.83443516377033672813 \dots, \quad \mathcal{R} = 0.64590122434782605589 \dots,$$

and the corresponding Gaussian coefficient. The rational coefficients in (6.9) and the derivatives in (6.23) are checked exactly.

N	Quartic: $D_N N^{5/4}/C_4$	Ordinary: $D_N N^{3/2}/(C_2 R^N)$
50	1.020979292	1.011642413
100	1.015637935	1.005539661
200	1.011457824	1.002703122
500	1.00747044	1.001065805
1000	1.005362011	1.000530376
2000	1.003831271	1.000264561

Table 3: Ratios predicted to approach one by [theorem 6.1](#). The quartic column uses the symmetric boundary; the ordinary column uses $A = 0.2$, $B = 0.1$.

The archive contains the exact source, numerical result files, figure-generation code, three vector figures with raster previews, and build instructions. The checked dependency versions are recorded in the verification README. Regenerating the numerical illustrations is optional for compiling the article because the generated figure and table files are included.

(τ, η)	N	Normalized sum	Ratio to limiting integral
$(0.4, 0.7)$	100	0.7504156763	1.015553592
	500	0.744372732	1.007375546
	2000	0.7417091222	1.003770826
$(-0.5, 0)$	100	0.4244297765	1.014013046
	500	0.4213501263	1.006655397
	2000	0.4199907618	1.003407714
$(0, 1)$	100	0.5883845562	1.019112524
	500	0.5825439869	1.008996355
	2000	0.5799991133	1.004588502

Table 4: The normalized quantity in (6.18) and its ratio to $\mathcal{P}(\tau, \eta)$ for three parameter pairs. The checks use the leading profile; the first correction and uniform remainder are proved analytically in section 6.3.

A Repository provenance, comparison, and scope of the audit

A.1 Reproducible source snapshot

The comparison uses the repository at [VladimirReshetnikov/ProveIt](#), commit

8e9cd6f00e0ac79c4425ff6abdfa64b497cf3f41.

The current transseries reports were inspected under

Analysis/Transseries/docs/series-and-transseries/.

The local inventory comprised 218 TeX and Markdown files in the transseries subtree. The canonical TeX source at that snapshot has 56,142 lines; the comparison used targeted searches and theorem-level reads, not a claim to have independently validated its entire contents. External literature was checked through the primary sources listed below.

A.2 What the repository already contains

Report	Relevant existing content and boundary
Support-controlled reversion [2]	Support-controlled Lagrange inversion, exact-core normalization, and restrictions on preserving a fixed one-exponential alphabet.
Nonlinear Stokes transport [3]	Finite complex mixed-action coefficients, exact action collisions, and weighted countable analytic perturbations. Its Section 3 is the starting coefficient identity for the present complex models.
Resonance-block summation [4]	Positive-action support and blockwise realization. Its research programme requests complex cone conditions and inverse-resurgence information with close primitive poles.
Negative-ray summation [5]	The inverse Borel formula and the coefficient majorant $e^{\tau r} \sum_{n \geq 1} M^n r^{2n-2} / (n!(n-1)!)$, including the type estimate $\tau + 2\sqrt{M}$. The present article identifies the Bessel function explicitly, proves sharpness, and supplies matching inverse-domain constants.

Report	Relevant existing content and boundary
Critical inverse reports [6, 7]	Critical charts for specified real positive-action classes; these do not establish unrestricted complex critical closure.
Finite-jet pressure laws [8]	Pressure asymptotics for real positive factorial-tail weights. This is related asymptotic methodology, but a different model from the orthogonal-action amplitude-convergence boundary.

In the pinned resonance-block source, the complex-action geometry question occurs at source lines 1621–1627, and the inverse-resurgence question at lines 1583–1591. The finite complex mixed-action identity in the nonlinear Stokes source occurs at lines 254–285. The earlier inverse Borel majorant in the negative-ray source occurs at lines 870–920. These locations are provenance aids for this snapshot, not stable line references to later repository versions.

A.3 The extension established here

The strict-cone theory resolves the geometric support part of the complex-action question for a finite alphabet, once the stated analytic summability and tail hypotheses are supplied. It distinguishes finite fibers from properness and proves a direction-independent completion, dense full-support sorting walls, and finite atlases at any prescribed exponential accuracy.

The exact two-action convergence domain, balanced quartic degree law, $4/3$ logarithmic boundary law, and angular crossover were not found in the targeted search of the 218 local source files. The repository’s finite-jet pressure report concerns a different factorial-tail problem. This comparison supports describing [section 6](#) as an extension of the inspected repository treatment; it does not establish world-first priority.

The sine-kernel analysis resolves a specific uniformity problem off the real Borel directions and supplies filtered continuation by a verified application of a published theorem. It does not determine every inverse singular germ, give a uniform constant on all sheets, construct arbitrary complex resonance blocks, or prove closure of all possible complex transseries. The research programme identifies those remaining tasks explicitly.

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- [4] VladimirReshetnikov/ProveIt contributors. Report on resonance-block summation of transseries. [Pinned TeX source](#).
- [5] VladimirReshetnikov/ProveIt contributors. Report on negative-ray summation and exponential feedback. [Pinned TeX source](#).
- [6] VladimirReshetnikov/ProveIt contributors. Report on critical transseries and a moving fold. [Pinned TeX source](#).

- [7] VladimirReshetnikov/ProveIt contributors. Report on critical Hahn transseries beyond finite-action folds. [Pinned TeX source](#).
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